RESEARCH ARTICLE

Do Online Product Listings Accurately Describe What Arrives in the Package?

Abstract

Online shoppers must decide whether to buy from a written listing before they can inspect the product, yet how well listings match what actually ships is rarely measured directly. We audited a stratified sample of marketplace listings across several product categories and fulfillment arrangements by purchasing each item and having blinded coders compare it against the specific claims its listing made. A substantial minority of listings failed at least one of their own claims, and inaccuracy varied systematically with how a listing was fulfilled and with the product category. Audited inaccuracy predicted buyers' later item-not-as-described complaints, but it bore little relation to the listing's star rating. Descriptive accuracy can therefore be measured independently of buyer sentiment, and the reputation signals platforms rely on are weak proxies for whether a listing tells the truth about the product.

Significance. Marketplaces and regulators increasingly ask whether product information can be trusted, but the metrics available to them—star ratings and aggregate complaint counts—mix accuracy with everything else a buyer reacts to. Auditing listings against their own claims yields a direct, buyer-independent measure of descriptive accuracy and shows where inaccuracy concentrates. Separating "is the description true" from "were buyers satisfied" matters for platform enforcement, consumer-protection policy, and any attempt to hold particular kinds of sellers accountable.

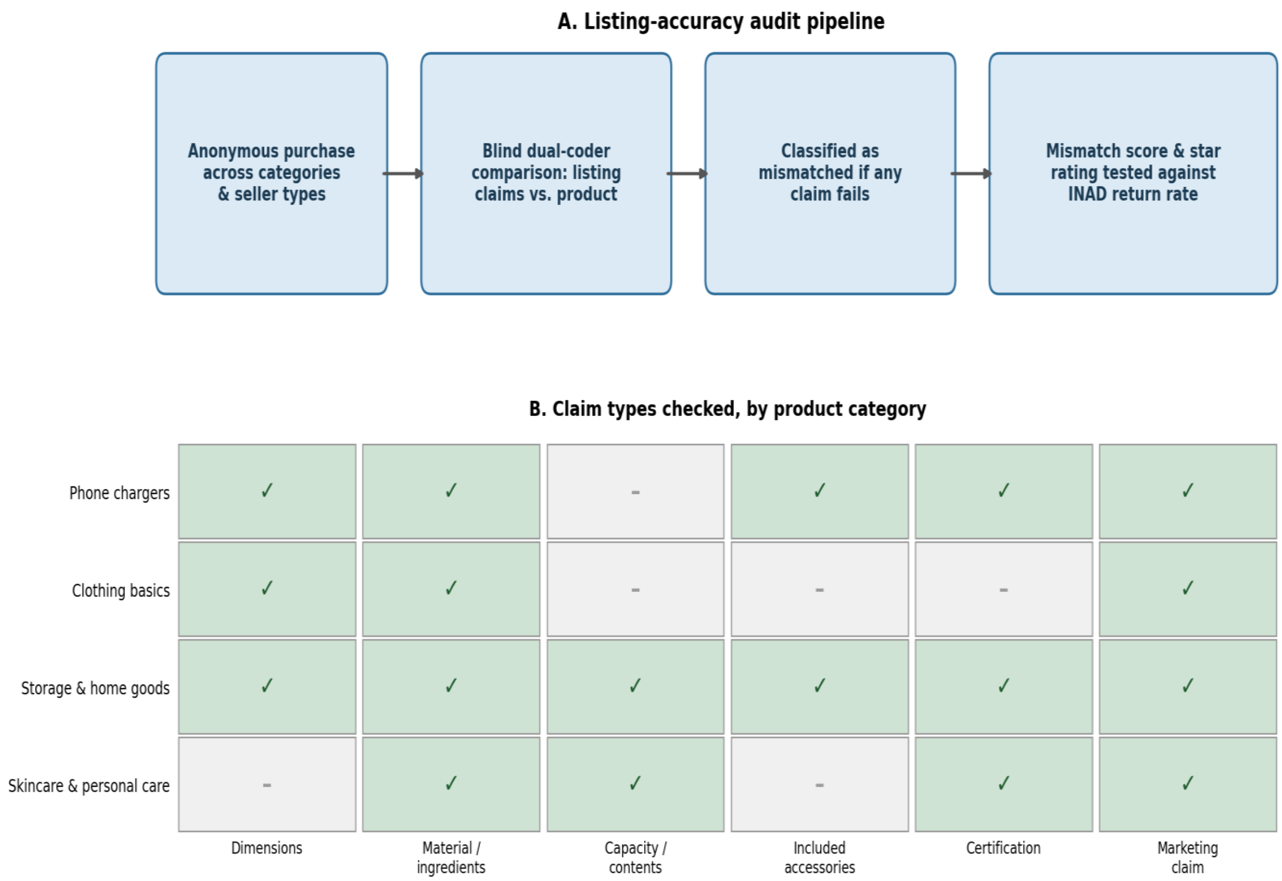

Figure 1. Listing-accuracy audit pipeline (A) and the claim types checked, by product category (B).

Introduction

Buying online is a decision made under uncertainty: the shopper commits money on the strength of a written description before the physical product can be examined (Akerlof, 1970). Some attributes remain hard to verify even after delivery, but many are plainly checkable once the box is open—an item's dimensions, its capacity, the materials it is made of (Nelson, 1970). Listings routinely assert such checkable facts, yet whether those specific assertions hold up against the delivered object has seldom been tested head-on.

Platforms bridge this gap with reputation systems—star ratings and reviews meant to stand in for direct inspection (Dellarocas, 2003; Chevalier & Mayzlin, 2006). But a rating is an aggregate signal that blends price, delivery, service, and expectation with product quality, and it is vulnerable to selection and manipulation (Hu et al., 2009; Mayzlin et al., 2014). A high rating therefore need not mean an accurate description, and whether ratings track descriptive accuracy in particular is an open question.

How a listing is fulfilled may also matter. Large marketplaces mix first-party retail, in which the platform is the seller (platform-sold); third-party sellers whose goods the platform warehouses and ships (fulfilled-by-platform); and third-party sellers who ship the item themselves (seller-shipped). These arrangements differ in how much the platform handles the physical product and in the incentives and selection of the sellers behind them (Nosko & Tadelis, 2015; Einav et al., 2016), so accuracy might vary with who actually stands behind the package. Product categories, in turn, may differ in which claims are easiest to get wrong.

We asked three questions: how often listings fail their own claims; whether inaccuracy varies with fulfillment type and product category; and whether audited inaccuracy lines up with the two buyer-facing signals a platform can see—complaint-driven returns and star ratings. We treated the audited mismatch as the primary measure and the two buyer-facing signals as secondary outcomes to be predicted, expecting accuracy to relate more closely to concrete complaints than to overall ratings. We did not set out to establish how large a discrepancy buyers are willing to accept before rejecting a product.

Method

Sampling and purchases. *We* drew a stratified random sample of active listings from a large general marketplace, crossing four product categories—skincare, phone chargers, clothing, and storage goods—with the three fulfillment types. In total 480 listings were sampled, balanced across the 12 category × fulfillment cells, and each sampled listing was purchased once and shipped to the lab; the listing, not the individual order, was the unit of sampling and of analysis. The categories were chosen to span quickly checkable and harder-to-verify attributes and to let us compare both how often and in what way listings were inaccurate; the three fulfillment types were included to test whether platform oversight of the physical item relates to accuracy.

Accuracy coding. Two trained coders, blind to fulfillment type, price, and the listing's rating, compared each delivered item against the claims that its own listing made, using a fixed rubric of checkable claim types: dimensions, weight, materials or ingredients, capacity or contents, color, quantity, and category-specific functional specifications (for chargers, rated output). Each applicable claim was scored pass or fail; a listing was classified as a mismatch if it failed any one of its applicable claims. We also recorded a continuous mismatch score, the share of a listing's applicable claims that failed. Coders agreed on the large majority of individual claims, and disagreements were resolved by discussion.

Outcomes and analysis. The primary outcome was the listing-level mismatch classification. As secondary outcomes, each listing's mismatch score was linked to two buyer-facing signals obtained from the platform: its item-not-as-described (INAD) return rate and its mean star rating. Fulfillment type and category varied between listings, whereas claim type varied within a listing. Mismatch was modeled with mixed-effects logistic regression including fulfillment type, category, and their interaction, with random intercepts for seller and listing. We then tested whether mismatch scores predicted INAD returns and ratings.

Results

About a third of listings failed at least one of their own claims. Mismatch rose consistently from platform-sold to fulfilled-by-platform to seller-shipped listings, and this ordering held in every category (Fig. 2; Table 1). Because the fulfillment types never changed rank across products, the seller-type difference was a stable main effect rather than something confined to particular categories.

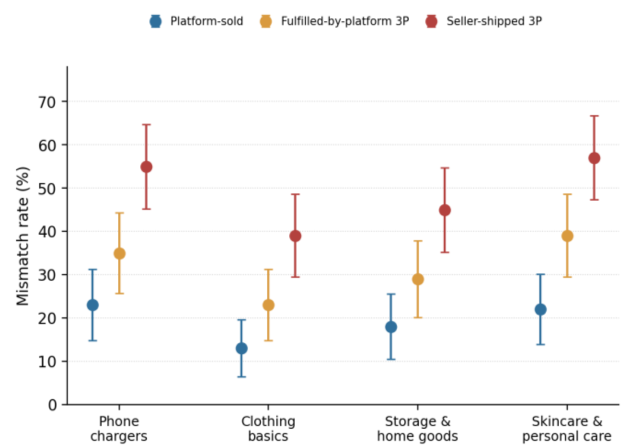

Figure 2. *Mismatch rate by category and seller type. Error bars are 95% CIs.*

Categories differed both in how often and in how they were inaccurate. Skincare and phone chargers had the highest mismatch rates overall, clothing and storage goods the lowest. More telling, the dominant failed claim differed by category—ingredient and content claims for skincare, rated-output and capacity claims for chargers, dimension claims for clothing, and capacity-or-contents claims for storage goods (Table 1). The same fulfillment ordering was thus expressed through different specific inaccuracies depending on the product, so knowing that seller-shipped listings are least accurate does not by itself reveal what will be wrong.

The audited mismatch score tracked buyers' own complaints: listings that failed more of their claims had higher INAD return rates (Fig. 3B). It did not track star ratings, which stayed high and nearly flat across the full range of mismatch (Fig. 3A). Descriptive accuracy and reputational standing came apart—an inaccurate listing could still carry an unremarkable or even favorable rating—so the two signals a platform sees are not interchangeable measures of whether a listing is truthful.

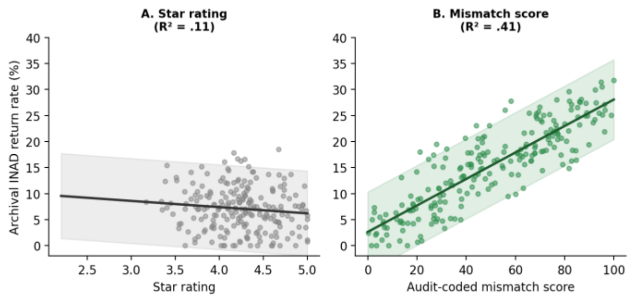

Figure 3. *Predicting archival item-not-as-described return rate from star rating (A) vs. audit-coded mismatch score (B), with 95% CI bands.*

Category	Platform-sold	Fulfilled-by-platform	Seller-shipped	Most common failed claim
Skincare	24%	38%	52%	Materials / ingredients
Phone chargers	22%	34%	48%	Rated output / capacity
Clothing	14%	24%	34%	Dimensions
Storage goods	12%	22%	33%	Capacity / contents

Table 1. *Percentage of listings that failed at least one of their own claims, by product category and fulfillment type, with the most common failed claim in each category.*

Conclusion

Listings frequently misdescribe the delivered product, and they do so unevenly: accuracy falls as the platform's hold on the physical item loosens, and both the rate and the character of inaccuracy depend on the category. Because the audit compared each item to its own claims rather than to a category ideal, these are listings failing their specific promises; and because audited mismatch predicted independent complaint behavior, the measure captures something buyers themselves act on.

That star ratings barely moved with accuracy is the study's most consequential qualification. We read it as a sign that ratings reflect overall purchase satisfaction—price, delivery, service, and met or unmet expectations—rather than descriptive accuracy in particular, which makes them ill-suited as an accuracy signal even though they remain the buyer's most visible cue.

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